

Boost shadow-equivalence report

Method 1 (shadow execution) of arXiv 2606.13882v1, on Boost's Nightscout devicestatus. Window: last 30 days. Users anonymised. Generated by `shadow_equivalence.py`.

What this measures (and what it doesn't)

Boost logs paired per-cycle outputs, so we can compute the paper's per-component agreement directly:

- **determineBasal (SMB): a DIVERGENCE metric.** V1 (acting) vs V5 — these are *different* algorithms, so divergence is intentional. The mismatch rate quantifies how different V5 is from V1 and where — the rigorous version of beta-readiness. (V1 SMB = units on shadow cycles; the V1 would= counterfactual on V5-active cycles. Tolerance $\pm 0.05U$.)
- **ISF EMA overlay: an EQUIVALENCE metric.** `isfShadow_deltaPct` is the V4.4.2 EMA($\tau=3h$) sensitivity overlay vs the instantaneous ratio actually used — same algorithm, two computations. Should be small/bounded ($\pm 5\%$).
- **NOT covered:** glucose-outcome clinical equivalence (Parkes Error Grid on counterfactual glucose) needs the replay harness (Method 2 — the V6 Phase-0 `replay.py`), not yet built. This is decision divergence, not simulated outcomes.

determineBasal — V1 vs V5 SMB divergence

user	days	cycles	agree $\leq 0.05U$	V5 doses more	V5 doses less	p95 $ \Delta $	mean Δ	net V1→V5
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30	13,274	80%	9%	11%	0.45U	-0.018U	933→698U	A
24	8,589	74%	12%	14%	1.05U	-0.022U	1485→1296U	B
23	8,185	63%	14%	23%	0.85U	+0.005U	1464→1506U	C
13	5,380	79%	15%	6%	0.45U	+0.021U	320→434U	D
11	4,138	74%	13%	13%	0.45U	+0.008U	415→449U	

High agreement = V5 mostly matches V1 (most cycles neither doses); the divergence concentrates in the meal cycles. 'p95 $|\Delta|$ ' is the 95th-percentile absolute per-cycle SMB difference — the size of the biggest routine disagreements.

ISF EMA-overlay — equivalence

user	cycles w/ ISF shadow	within $\pm 5\%$	p95 $ \Delta\% $	mean $\Delta\%$	--- ---	---	---	self
2,951	57%	23.9%	+7.8%	A	8,589	33%	32.0%	+5.7%
B	8,175	28%	38.8%	+1.7%	C	5,380	58%	16.8%
D	0	—	—	—				

If the EMA overlay is within $\pm 5\%$ on ~all cycles it's clinically equivalent to instant ISF (safe to adopt as a 'computational/heuristic' change per the paper's taxonomy); large/ systematic deltas mean it's a behavioural change needing gradual transition.

meal-state distribution

- **self:** IDLE 56%, OBSERVING 26%, COMMITTED 8%, RECOVERING 7%, CONFIRMED 4%
- **A:** IDLE 64%, OBSERVING 24%, RECOVERING 5%, CONFIRMED 4%, COMMITTED 3%
- **B:** IDLE 56%, OBSERVING 26%, RECOVERING 8%, COMMITTED 5%, CONFIRMED 4%
- **C:** IDLE 62%, OBSERVING 27%, RECOVERING 5%, CONFIRMED 3%, COMMITTED 2%
- **D:** IDLE 72%, OBSERVING 22%, RECOVERING 3%, CONFIRMED 2%, COMMITTED 2%

Reading it (bug taxonomy — paper Part I)

- **Factual** bugs (objective, fix immediately): already done — HypoCaution inversion, getGlucoseStatusData-null, constraint gaps.
- **Heuristic** (co-adapted with the user — transition GRADUALLY): V5 aggression/dose calibration, the fast-carb confirm timing. The SMB-divergence columns above quantify how far V5 has moved from the V1 the user co-adapted to — large moves argue for shadow-first + gradual.
- **Computational** (numeric equivalence): cross-repo port (3.4 Kotlin → V4 Compose). Validate with the SMB + ISF agreement metrics here once the V4 build emits shadow data.

Next: Method 2 (replay) for true clinical equivalence

Build `replay.py` (V6 Phase 0): re-run `determine_basal` over historical NS inputs under two algorithm versions, then Parkes-Error-Grid the resulting dosing/predicted-glucose. That gives the paper's clinical-equivalence gate ($\geq 99\%$ Parkes A/B, TIR within tolerance) — the acceptance test for the fast-carb fix, the V5 beta decision, and the V4 port.